QUBO Formulation: Cardinality-Constrained Portfolio Selection Problem

Cardinality-Constrained Portfolio Selection Problem

Problem Statement. Given a set of N assets, each asset i is associated with an expected return score r_i. Each distinct pair of assets (i, j) is associated with a covariance or risk penalty c_{ij}. The task is to select exactly K assets to form a portfolio. The objective is to maximize the risk-adjusted score, defined as the sum of the expected returns of the selected assets minus the sum of the pairwise risk penalties between the selected assets.

Decision Variables. For each asset $i \in \{0, 1, \dots, N-1\}$, we introduce a binary decision variable $x_i \in \{0, 1\}$. The variable $x_i = 1$ indicates that asset i is included in the portfolio, while $x_i = 0$ indicates that it is excluded.

QUBO Derivation. **Step 1.** The original problem seeks to maximize the risk-adjusted score. To conform to the standard QUBO minimization convention, we negate the objective function:

$$\min_x \left(-\sum_{i=0}^{N-1} r_i x_i + \sum_{0 \le i < j \le N-1} c_{ij} x_i x_j \right). \tag{1}$$

Step 2. The cardinality constraint requires that exactly K assets are selected:

$$\sum_{i=0}^{N-1} x_i = K. \tag{2}$$

Step 3. We enforce the constraint via a quadratic penalty term with coefficient $P > 0$:

$$\mathcal{P}(x) = P \left(\sum_{i=0}^{N-1} x_i - K \right)^2. \tag{3}$$

Expanding the square and applying the idempotent property $x_i^2 = x_i$ for binary variables yields:

$$\left(\sum_{i=0}^{N-1} x_i - K\right)^2 = \sum_{i=0}^{N-1} x_i^2 + 2 \sum_{0 \le i < j \le N-1} x_i x_j - 2K \sum_{i=0}^{N-1} x_i + K^2 \tag{4}$$

$$= \sum_{i=0}^{N-1} (1-2K) x_i + 2 \sum_{0 \le i < j \le N-1} x_i x_j + K^2. \tag{5}$$

Multiplying by P gives the general penalty contribution:

$$\mathcal{P}(x) = P \sum_{i=0}^{N-1} (1-2K) x_i + 2P \sum_{0 \le i < j \le N-1} x_i x_j + PK^2. \tag{6}$$

Step 4. Combining the negated objective and the penalty term yields the total QUBO Hamiltonian $H(x)$:

$$H(x) = \sum_{i=0}^{N-1} \left(P(1-2K) - r_i\right) x_i + \sum_{0 \le i < j \le N-1} \left(c_{ij} + 2P\right) x_i x_j + PK^2. \tag{7}$$

Step 5. Reading off the coefficients from $H(x)$, the upper-triangular QUBO matrix entries $Q_{i,j}$ and the constant offset c are given by:

$$Q_{i,i} = P(1-2K) - r_i, \qquad \text{for } i \in \{0, \ldots, N-1\}, \tag{8}$$

$$Q_{i,j} = c_{ij} + 2P, \qquad \text{for } 0 \le i < j \le N-1, \tag{9}$$

$$Q_{i,j} = 0, \qquad \text{for } i > j, \tag{10}$$

$$c = PK^2. \tag{11}$$

Penalty Weight Justification. The penalty coefficient P must be chosen sufficiently large to ensure that any violation of the cardinality constraint incurs a higher energy cost than the maximum possible improvement in the objective function. A safe lower bound is $P > \max_i |r_i| + \sum_{0 \le i < j \le N-1} |c_{ij}|$. This guarantees that the penalty term strictly dominates the objective term for any infeasible assignment, thereby forcing the global minimum to reside within the feasible region.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^N$ recovers the optimal portfolio because (i) all feasible assignments satisfying $\sum_i x_i = K$ yield a penalty term of exactly zero, leaving only the original risk-adjusted objective to be minimized; and (ii) any infeasible assignment incurs a penalty strictly greater than the maximum possible gain achievable by altering the objective function. Consequently, the global minimum of $H(x)$ must correspond to a feasible assignment that optimally balances expected returns and risk penalties.

Illustrative Example. Consider a concrete instance with $N = 3$ assets and a target cardinality $K = 2$. The parameters are r_0, r_1, r_2 for expected returns and c_{01}, c_{02}, c_{12} for pairwise risk penalties. Substituting $K = 2$ into the general formulas, the concrete objective becomes:

$$\min_x \left(-r_0 x_0 - r_1 x_1 - r_2 x_2 + c_{01} x_0 x_1 + c_{02} x_0 x_2 + c_{12} x_1 x_2\right). \tag{12}$$

The penalty expansion yields linear coefficients $P(1-4) = -3P$, quadratic coefficients $2P$, and an offset $4P$. The resulting Hamiltonian is:

$$\begin{aligned} H(x) = (-3P - r_0)x_0 + (-3P - r_1)x_1 + (-3P - r_2)x_2 \\ + (c_{01} + 2P)x_0 x_1 + (c_{02} + 2P)x_0 x_2 + (c_{12} + 2P)x_1 x_2 + 4P. \end{aligned} \tag{13}$$

The corresponding QUBO matrix entries and offset are:

$$Q_{0,0} = -3P - r_0, \qquad Q_{1,1} = -3P - r_1, \qquad Q_{2,2} = -3P - r_2, \tag{14}$$

$$Q_{0,1} = c_{01} + 2P, \qquad Q_{0,2} = c_{02} + 2P, \qquad Q_{1,2} = c_{12} + 2P, \tag{15}$$

$$c = 4P. \tag{16}$$

Since $K = 2$, there are exactly three feasible assignments: $x^{(1)} = (1, 1, 0)$, $x^{(2)} = (1, 0, 1)$, and $x^{(3)} = (0, 1, 1)$. Evaluating $H(x)$ at these points (where the penalty term vanishes) gives:

$$H(x^{(1)}) = -r_0 - r_1 + c_{01} + 4P, \tag{17}$$

$$H(x^{(2)}) = -r_0 - r_2 + c_{02} + 4P, \tag{18}$$

$$H(x^{(3)}) = -r_1 - r_2 + c_{12} + 4P. \tag{19}$$

The optimal portfolio corresponds to the assignment x^* that minimizes $H(x^*)$. For instance, if $H(x^{(1)})$ is the minimum among the three, the optimal solution is $x^* = (1, 1, 0)$ with energy $H(x^*) = -r_0 - r_1 + c_{01} + 4P$. The maximum risk-adjusted score for this instance is recovered as $-(H(x^*) - c) = r_0 + r_1 - c_{01}$.